

tqb3h0qto

July 2, 2024

## 1 Bank customer churn analysis

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## 2 Problem statement:

In the rapidly evolving banking sector, customer retention has become a critical concern. Banks are increasingly seeking to understand the factors that influence customer decisions to stay with or leave their banking service provider. This project focuses on analyzing a dataset containing various attributes of bank customers to identify key predictors of customer churn. By leveraging data analytics, we aim to uncover patterns and insights that could help devise strategies to enhance customer retention and reduce churn rates.

---

```
[ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
from scipy.stats import ttest_1samp, ttest_ind, ttest_rel, f_oneway, chisquare,
↳chi2_contingency
from scipy.stats import shapiro # Shapiro-Wilk's test for Normality
from scipy.stats import levene # Levene's test for Equality of Variance
from scipy.stats import kruskal
from statsmodels.api import qqplot
```

```
[ ]: !wget "https://drive.google.com/file/d/1xh7DONDmxdg6IXTFzi_T-0c5D-GtI44W/view"
```

```
--2024-07-02 07:45:03-- https://drive.google.com/file/d/1xh7DONDmxdg6IXTFzi_T-
0c5D-GtI44W/view
Resolving drive.google.com (drive.google.com)... 142.250.98.138, 142.250.98.139,
142.250.98.101, ...
Connecting to drive.google.com (drive.google.com)|142.250.98.138|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
```

Saving to: 'view'

view [ <=> ] 90.01K --.-KB/s in 0.007s

2024-07-02 07:45:04 (12.5 MB/s) - 'view' saved [92168]

```
[ ]: df = pd.read_csv('/content/Bank-Records.csv')
df
```

```
[ ]:
  RowNumber  CustomerId  Surname  CreditScore  Geography  Gender  Age  \
0          1    15634602   Hargrave         619     France  Female  42
1          2    15647311     Hill         608      Spain  Female  41
2          3    15619304     Onio         502     France  Female  42
3          4    15701354     Boni         699     France  Female  39
4          5    15737888  Mitchell         850      Spain  Female  43
...      ...      ...      ...      ...      ...      ...
9995      9996    15606229   Obijiaku         771     France    Male  39
9996      9997    15569892  Johnstone         516     France    Male  35
9997      9998    15584532      Liu         709     France  Female  36
9998      9999    15682355  Sabbatini         772    Germany    Male  42
9999     10000    15628319    Walker         792     France  Female  28
```

```

  Tenure  Balance  NumOfProducts  HasCrCard  IsActiveMember  \
0        2      0.00             1          1              1
1        1  83807.86             1          0              1
2        8 159660.80             3          1              0
3        1      0.00             2          0              0
4        2 125510.82             1          1              1
...     ...     ...      ...      ...      ...
9995      5      0.00             2          1              0
9996     10  57369.61             1          1              1
9997      7      0.00             1          0              1
9998      3  75075.31             2          1              0
9999      4 130142.79             1          1              0
```

```

  EstimatedSalary  Exited  Complain  Satisfaction  Score  Card Type  \
0        101348.88       1         1              2    DIAMOND
1        112542.58       0         1              3    DIAMOND
2        113931.57       1         1              3    DIAMOND
3         93826.63       0         0              5      GOLD
4         79084.10       0         0              5      GOLD
...     ...     ...      ...      ...      ...
9995         96270.64       0         0              1    DIAMOND
9996        101699.77       0         0              5  PLATINUM
9997         42085.58       1         1              3    SILVER
9998         92888.52       1         1              2      GOLD
```

|      |          |   |   |   |         |
|------|----------|---|---|---|---------|
| 9999 | 38190.78 | 0 | 0 | 3 | DIAMOND |
|------|----------|---|---|---|---------|

|      | Point Earned |
|------|--------------|
| 0    | 464          |
| 1    | 456          |
| 2    | 377          |
| 3    | 350          |
| 4    | 425          |
| ...  | ...          |
| 9995 | 300          |
| 9996 | 771          |
| 9997 | 564          |
| 9998 | 339          |
| 9999 | 911          |

[10000 rows x 18 columns]

## 2.1 Exploratory Data Analysis

- Examine dataset structure, characteristics, and statistical summary.

```
[ ]: df.shape
```

```
[ ]: (10000, 18)
```

no of rows = 10000 no of cols = 18

```
[ ]: df.dtypes
```

```
[ ]: CreditScore          int64
      Geography           object
      Gender              object
      Age                 int64
      Tenure              int64
      Balance             float64
      NumOfProducts       int64
      HasCrCard           int64
      IsActiveMember      int64
      EstimatedSalary     float64
      Exited              int64
      Complain            int64
      Satisfaction Score  int64
      Card Type           object
      Point Earned        int64
      dtype: object
```

```
[ ]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   RowNumber              10000 non-null  int64
1   CustomerId             10000 non-null  int64
2   Surname                10000 non-null  object
3   CreditScore            10000 non-null  int64
4   Geography              10000 non-null  object
5   Gender                 10000 non-null  object
6   Age                   10000 non-null  int64
7   Tenure                 10000 non-null  int64
8   Balance                10000 non-null  float64
9   NumOfProducts          10000 non-null  int64
10  HasCrCard              10000 non-null  int64
11  IsActiveMember         10000 non-null  int64
12  EstimatedSalary        10000 non-null  float64
13  Exited                 10000 non-null  int64
14  Complain               10000 non-null  int64
15  Satisfaction Score     10000 non-null  int64
16  Card Type              10000 non-null  object
17  Point Earned           10000 non-null  int64
dtypes: float64(2), int64(12), object(4)
memory usage: 1.4+ MB

```

## 2.2 missing values

```
[ ]: df.isnull().sum()
```

```

[ ]: RowNumber          0
     CustomerId         0
     Surname            0
     CreditScore        0
     Geography          0
     Gender             0
     Age               0
     Tenure            0
     Balance           0
     NumOfProducts     0
     HasCrCard         0
     IsActiveMember    0
     EstimatedSalary   0
     Exited            0
     Complain          0
     Satisfaction Score 0
     Card Type         0

```

```
Point Earned          0
dtype: int64
```

insights \* From above analysis, we found that there are no missing values in any column.

```
[ ]: df.duplicated().sum() # checking for duplicate values
```

```
[ ]: 0
```

in our dataset we don't have duplicate values.

```
[ ]: df.describe()
```

```
[ ]:
      RowNumber  CustomerId  CreditScore      Age      Tenure \
count  10000.00000  1.000000e+04  10000.000000  10000.000000  10000.000000
mean     5000.50000  1.569094e+07   650.528800   38.921800    5.012800
std     2886.89568  7.193619e+04    96.653299   10.487806    2.892174
min         1.00000  1.556570e+07   350.000000   18.000000    0.000000
25%     2500.75000  1.562853e+07   584.000000   32.000000    3.000000
50%     5000.50000  1.569074e+07   652.000000   37.000000    5.000000
75%     7500.25000  1.575323e+07   718.000000   44.000000    7.000000
max    10000.00000  1.581569e+07   850.000000   92.000000   10.000000
```

```
      Balance  NumOfProducts  HasCrCard  IsActiveMember \
count  10000.000000  10000.000000  10000.000000  10000.000000
mean     76485.889288    1.530200    0.705500    0.515100
std     62397.405202    0.581654    0.455840    0.499797
min         0.000000    1.000000    0.000000    0.000000
25%         0.000000    1.000000    0.000000    0.000000
50%     97198.540000    1.000000    1.000000    1.000000
75%    127644.240000    2.000000    1.000000    1.000000
max    250898.090000    4.000000    1.000000    1.000000
```

```
      EstimatedSalary      Exited      Complain  Satisfaction Score \
count  10000.000000  10000.000000  10000.000000  10000.000000
mean     100090.239881    0.203800    0.204400    3.013800
std     57510.492818    0.402842    0.403283    1.405919
min         11.580000    0.000000    0.000000    1.000000
25%     51002.110000    0.000000    0.000000    2.000000
50%    100193.915000    0.000000    0.000000    3.000000
75%    149388.247500    0.000000    0.000000    4.000000
max    199992.480000    1.000000    1.000000    5.000000
```

```
      Point Earned
count  10000.000000
mean     606.515100
std     225.924839
min     119.000000
```

```

25%      410.000000
50%      605.000000
75%      801.000000
max      1000.000000

```

```
[ ]: df.columns
```

```
[ ]: Index(['RowNumber', 'CustomerId', 'Surname', 'CreditScore', 'Geography',
          'Gender', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'HasCrCard',
          'IsActiveMember', 'EstimatedSalary', 'Exited', 'Complain',
          'Satisfaction Score', 'Card Type', 'Point Earned'],
          dtype='object')
```

## 2.3 Dropping Irrelevant Features

```
[ ]: df = df.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1)
df
```

```
[ ]:
   CreditScore Geography Gender Age  Tenure  Balance  NumOfProducts \
0           619   France  Female  42     2     0.00             1
1           608    Spain  Female  41     1  83807.86             1
2           502   France  Female  42     8 159660.80             3
3           699   France  Female  39     1     0.00             2
4           850    Spain  Female  43     2 125510.82             1
...         ...      ...    ...   ...   ...      ...             ...
9995         771   France   Male  39     5     0.00             2
9996         516   France   Male  35    10  57369.61             1
9997         709   France  Female  36     7     0.00             1
9998         772  Germany   Male  42     3  75075.31             2
9999         792   France  Female  28     4 130142.79             1
```

```

   HasCrCard  IsActiveMember  EstimatedSalary  Exited  Complain \
0           1               1      101348.88       1         1
1           0               1      112542.58       0         1
2           1               0      113931.57       1         1
3           0               0       93826.63       0         0
4           1               1       79084.10       0         0
...         ...            ...            ...      ...      ...
9995         1               0       96270.64       0         0
9996         1               1      101699.77       0         0
9997         0               1       42085.58       1         1
9998         1               0       92888.52       1         1
9999         1               0       38190.78       0         0
```

```

   Satisfaction Score Card Type  Point Earned
0                   2   DIAMOND           464
1                   3   DIAMOND           456
```

|      |     |          |     |
|------|-----|----------|-----|
| 2    | 3   | DIAMOND  | 377 |
| 3    | 5   | GOLD     | 350 |
| 4    | 5   | GOLD     | 425 |
| ...  | ... | ...      | ... |
| 9995 | 1   | DIAMOND  | 300 |
| 9996 | 5   | PLATINUM | 771 |
| 9997 | 3   | SILVER   | 564 |
| 9998 | 2   | GOLD     | 339 |
| 9999 | 3   | DIAMOND  | 911 |

[10000 rows x 15 columns]

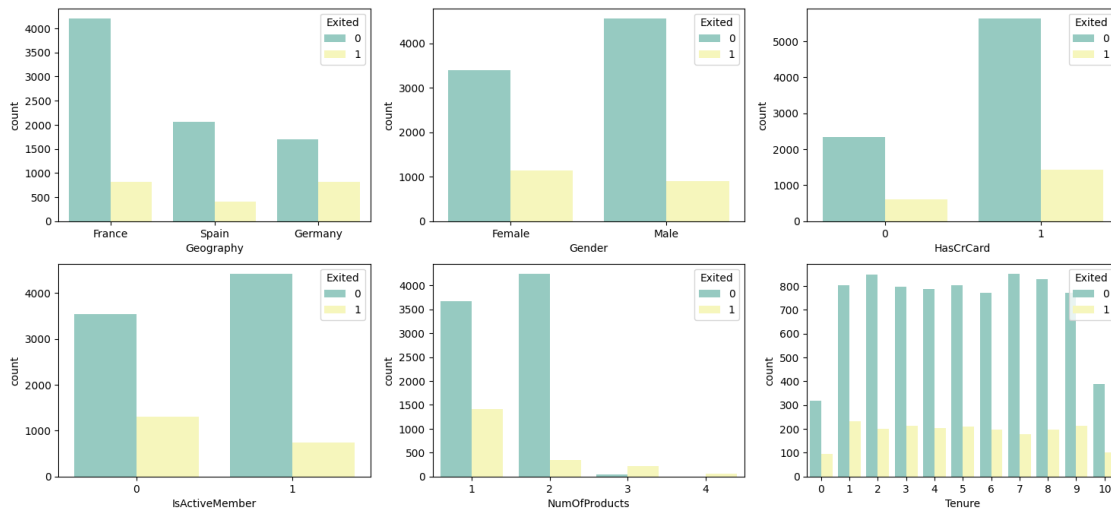
```
[ ]: df.shape
```

```
[ ]: (10000, 15)
```

## 2.4 Count plots to map the dependence of 'Exited' column on categorical variable.

```
[ ]: fig,ax = plt.subplots(2,3,figsize=(18,8))
sns.countplot(x='Geography',hue='Exited',data=df,ax=ax[0,0],palette='Set3')
sns.countplot(x='Gender',hue='Exited',data=df,ax=ax[0,1],palette='Set3')
sns.countplot(x='HasCrCard',hue='Exited',data=df,ax=ax[0,2],palette='Set3')
sns.countplot(x='IsActiveMember',hue='Exited',data=df,ax=ax[1,0],palette='Set3')
sns.countplot(x='NumOfProducts',hue='Exited',data=df,ax=ax[1,1],palette='Set3')
sns.countplot(x='Tenure',hue='Exited',data=df,ax=ax[1,2],palette='Set3')
```

```
[ ]: <Axes: xlabel='Tenure', ylabel='count'>
```



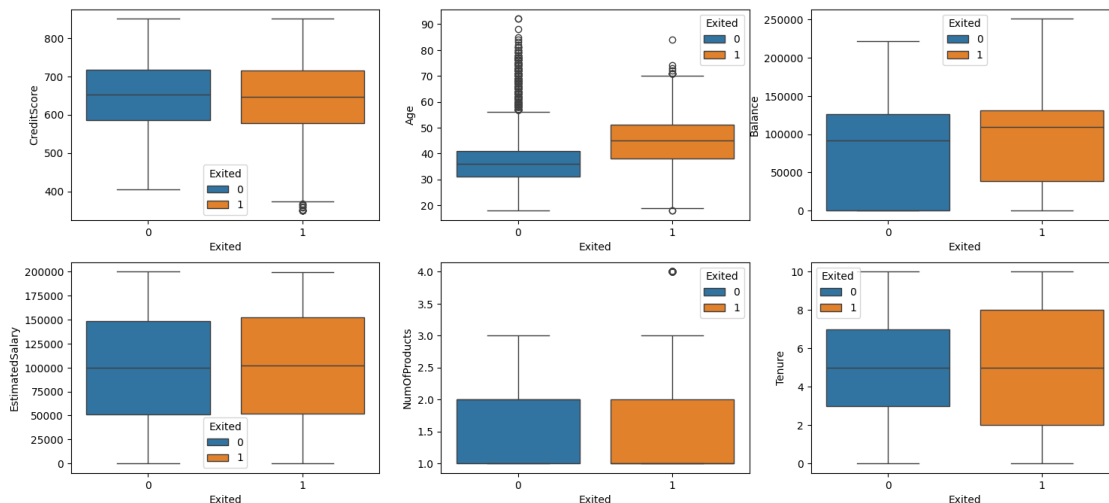
insights:

- **Geography:** We can see that majority of the data is about people France. Ideally for an evenly-distributed data, if the amount of people from a place is the majority, then the majority of churning should also be within that group. However, it is not so in this case as we see that number of exited people who belong to Germany is almost equal to the number of exits from France.
- **Gender:** We can clearly see the Female customers had more exits than the male customers.
- **Credit cards:** It is generally expected that people who have more interactions and products of the bank, would likely be retained for a longer time. However, we can see that people who have credit cards have more exits than those who do not own credit cards.
- **Active Member:** This is an expected observation. We can see that inactive members have been churned more than members who are active.
- **Number of Products:** This is also an expected observation, where we see that customers who own more products from the bank are likely to be retained for a longer time than those who own less products.
- **Tenure:** We see that the tenure of a customer does not really tell us much if that customer is likely to be churned or not.

```
[ ]: # Count column plots to map the dependence of 'Exited' column on continuous and
      ↪ numerical features
```

```
fig, ax = plt.subplots(2, 3, figsize=(18,8))
```

```
sns.boxplot(data=df, x='Exited', y='CreditScore', hue='Exited', ax=ax[0][0])
sns.boxplot(data=df, x='Exited', y='Age', hue='Exited', ax=ax[0][1])
sns.boxplot(data=df, x='Exited', y='Balance', hue='Exited', ax=ax[0][2])
sns.boxplot(data=df, x='Exited', y='EstimatedSalary', hue='Exited', ax=ax[1][0])
sns.boxplot(data=df, x='Exited', y='NumOfProducts', hue='Exited', ax=ax[1][1])
sns.boxplot(data=df, x='Exited', y='Tenure', hue='Exited', ax=ax[1][2])
plt.show()
```



- **Credit Score:** We can see that Credit Score does not have much effect on the customer churn.
- **Age:** Here we can see that the older customers are more likely to be churned from the bank. This is most probably to keep a younger manpower in the organization.
- **Balance:** When it comes to Balance, we see that the bank is losing a significant number of customers with high balance in their accounts. This is likely to affect the bank's capital as well.
- **Estimated Salary:** Estimated Salary does not seem to affect the customer churn much.
- **Number of Products:** We see that the number of products also does not seem to affect the customer churn.
- **Tenure:** For tenure, as we can see here too, customer belonging more to the two extreme tenure groups (new joiners and older ones) are more likely to be churned.

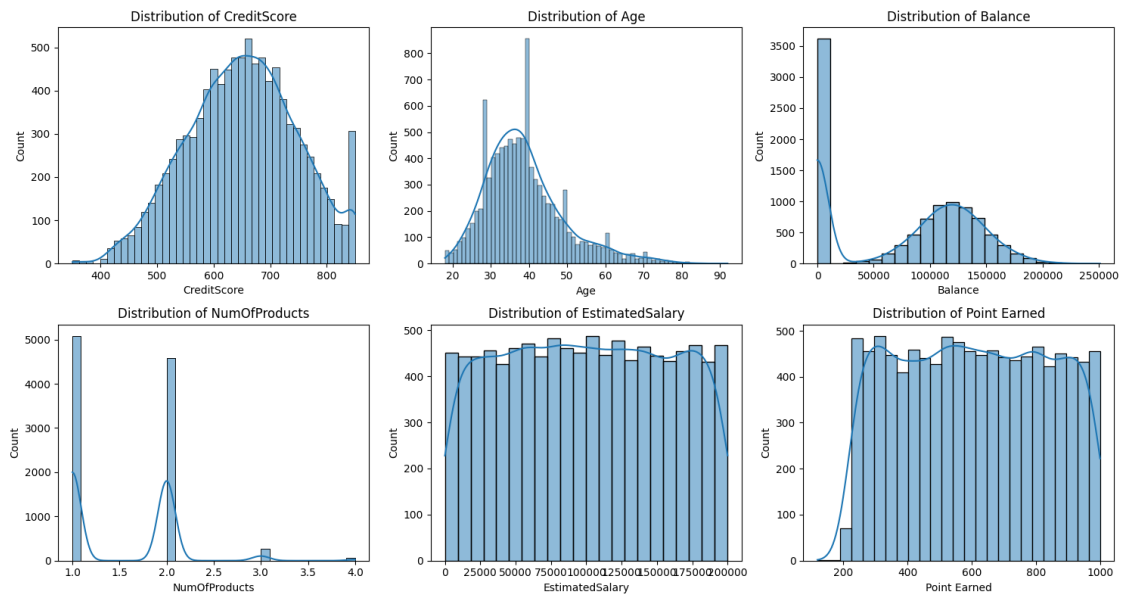
```
[ ]: Numerical_columns=['CreditScore','Age','Balance','NumOfProducts','EstimatedSalary','Point_
↳Earned']
for col in Numerical_columns:
    print(f'Mean of {col}: {df[col].mean()}')
    print(f'Median of {col}: {df[col].median()}')
    print(f'Mode of {col}: {df[col].mode()[0]}')
```

```
Mean of CreditScore: 650.5288
Median of CreditScore: 652.0
Mode of CreditScore: 850
Mean of Age: 38.9218
Median of Age: 37.0
Mode of Age: 37
Mean of Balance: 76485.889288
Median of Balance: 97198.54000000001
Mode of Balance: 0.0
Mean of NumOfProducts: 1.5302
Median of NumOfProducts: 1.0
Mode of NumOfProducts: 1
Mean of EstimatedSalary: 100090.239881
Median of EstimatedSalary: 100193.915
Mode of EstimatedSalary: 24924.92
Mean of Point Earned: 606.5151
Median of Point Earned: 605.0
Mode of Point Earned: 408
```

Analyze the distribution of key numerical variables using histograms and box plots to understand the spread and central tendency.

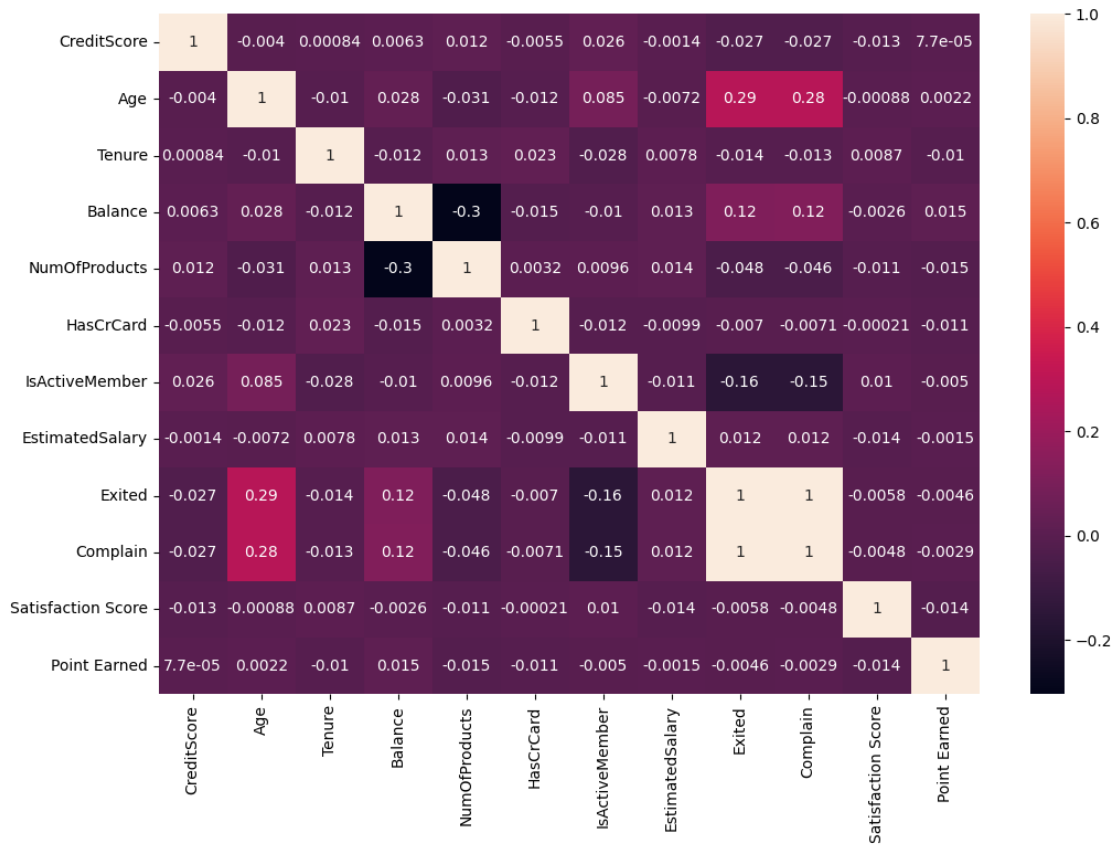
```
[ ]: plt.figure(figsize=(15, 8))
for i, col in enumerate(Numerical_columns):
```

```
plt.subplot(2, 3, i+1)
sns.histplot(df[col],kde=True)
plt.title(f'Distribution of {col}')
plt.xlabel(col)
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



##Let's see how the data correlates with each other

```
[66]: plt.figure(figsize=(12,8))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```



- It seems that our data does not correlate well with each other. The only one with a low correlation is 'Age' and 'Exited' this is because people who Exit happen to have a high chance to do so only between a particular age range.
- This lack of correlation could really be because there is truly no correlation between the variables or because of the method that is used to calculate the correlation which only measures the linear relationships between them

```
[ ]: data_gen=pd.crosstab(df['Gender'],df['Exited'],normalize='index')
data_gen
```

```
[ ]: Exited      0      1
Gender
Female  0.749285  0.250715
Male    0.835257  0.164743
```

```
[ ]: # Ho:There is no significant difference between churn rates of both
      ↪genders,males and females
      # Ha:There is a significant difference between churn rates of both
      ↪genders,males and females
```

```

crosstab_gen = pd.crosstab(df['Gender'], df['Exited'])
chi2, p, dof, expected = chi2_contingency(crosstab_gen)

# Print the results of the chi-square test
print("Chi-square test statistic:", chi2)
print("P-value:", p)

# Interpret the results
if p < 0.05:
    print("There is a statistically significant difference in churn rates_
    ↳between genders.")
else:
    print("There is no statistically significant difference in churn rates_
    ↳between genders.")

```

Chi-square test statistic: 112.39655374778587

P-value: 2.9253677618642e-26

There is a statistically significant difference in churn rates between genders.

### 3 Behavioral Analysis

```

[ ]: crosstab_prod=pd.crosstab(df['NumOfProducts'],df['Exited'])
crosstab_prod

```

```

[ ]: Exited          0      1
NumOfProducts
1              3675  1409
2              4241   349
3               46   220
4               0    60

```

```

[ ]: # Ho:There is no significant difference between churn rates of different No. of_
    ↳Products.
# Ha:There is a significant difference between churn rates of different No. of_
    ↳Products.

chi2, p, dof, expected = chi2_contingency(crosstab_prod)

# Print the results of the chi-square test
print("Chi-square test statistic:", chi2)
print("P-value:", p)

# Interpret the results
if p < 0.05:
    print("There is a statistically significant difference in churn rates_
    ↳between different No. of Products.")

```

```

else:
    print("There is no statistically significant difference in churn rates_
↪between No. of Products.")

```

Chi-square test statistic: 1501.5048306588592

P-value: 0.0

There is a statistically significant difference in churn rates between different No. of Products.

```

[ ]: crosstab_IsActive = pd.crosstab(df['IsActiveMember'],df['Exited'])
crosstab_IsActive

```

```

[ ]: Exited          0      1
      IsActiveMember
0          3546   1303
1          4416    735

```

```

[ ]: # Ho:There is no significant effect of IsActiveMember on Churning rates.
      # Ha:There is a significant effect of IsActiveMember on Churning rates.
      chi2, p, dof, expected = chi2_contingency(crosstab_IsActive)

      # Print the results of the chi-square test
      print("Chi-square test statistic:", chi2)
      print("P-value:", p)

      # Interpret the results
      if p < 0.05:
          print("Reject the Null Hypothesis")
      else:
          print("Fail to Reject the Null Hypothesis")

```

Chi-square test statistic: 243.6948024819593

P-value: 6.153167438113408e-55

Reject the Null Hypothesis

Active members are significantly less likely to churn. Engaging customers and promoting active use of services can improve retention.

## 4 Financial Analysis

```

[ ]: df.groupby('Exited')['Balance'].agg(['mean', 'median'])

```

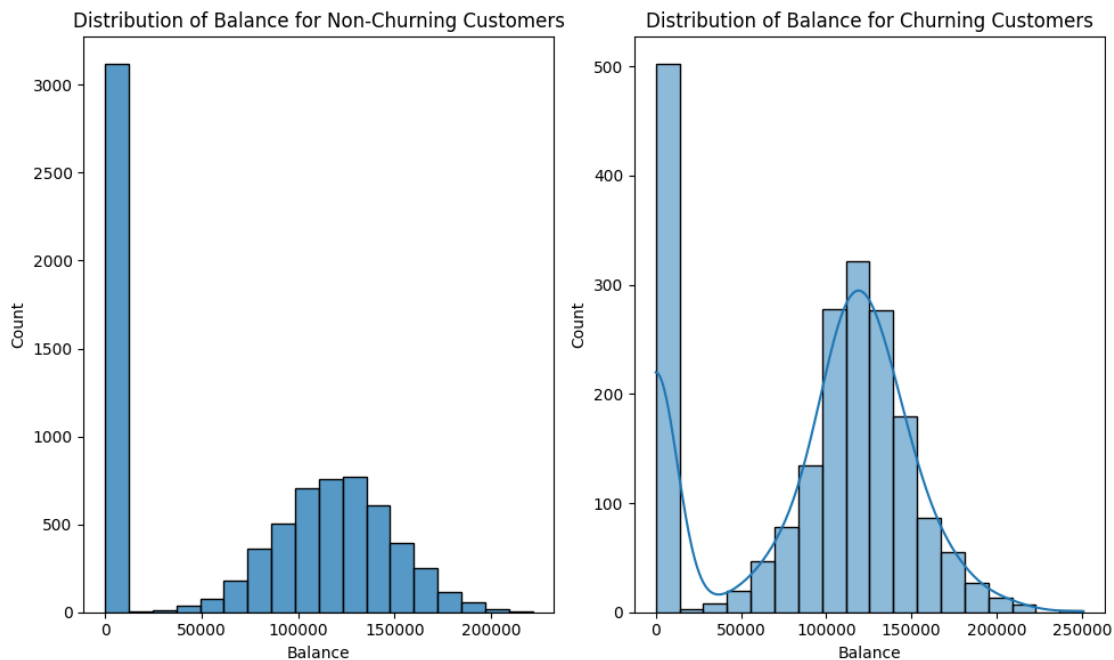
```

[ ]:
      mean      median
Exited
0      72742.750663   92063.06
1      91109.476006  109344.23

```

```
[69]: data_0=df[df['Exited']==0]
data_1=df[df['Exited']==1]
plt.figure(figsize=(10, 6))
plt.subplot(1, 2, 1)
sns.histplot(data_0['Balance'])
plt.title('Distribution of Balance for Non-Churning Customers')
plt.xlabel('Balance')
plt.ylabel('Count')

plt.subplot(1, 2, 2)
sns.histplot(data_1['Balance'],kde = True)
plt.title('Distribution of Balance for Churning Customers')
plt.xlabel('Balance')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



```
[ ]: # Ho:There is no significant effect of Balance on Churning rates.
# Ha:There is a significant effect of Balance on Churning rates.

from scipy.stats import ttest_ind
stats,p_val=ttest_ind(data_0['Balance'],data_1['Balance'])
print("tstats:",stats)
print("p_val:",p_val)
if p_val<0.05:
```

```

    print('Reject Null Hypothesis')
else:
    print('Fail to reject Null Hypothesis')

```

```

tstats: -11.940747722508185
p_val: 1.2092076077156017e-32
Reject Null Hypothesis

```

```

[ ]: crosstab_cc=pd.crosstab(df['HasCrCard'],df['Exited'],normalize='index')
crosstab_cc

```

```

[ ]: Exited          0          1
HasCrCard
0          0.791851  0.208149
1          0.798016  0.201984

```

```

[ ]: # Ho:There is no significant effect of Having a credit card on Churning rates.
# Ha:There is a significant effect of Having a credit card on Churning rates.

chi2, p, dof, expected = chi2_contingency(pd.
↳crosstab(df['HasCrCard'],df['Exited']))

# Print the results of the chi-square test
print("Chi-square test statistic:", chi2)
print("P-value:", p)

# Interpret the results
if p < 0.05:
    print('Reject the Null Hypothesis')
else:
    print("Fail to Reject the Null Hypothesis")

```

```

Chi-square test statistic: 0.4494039375253385
P-value: 0.5026181509009862
Fail to Reject the Null Hypothesis

```

## 5 Customer Satisfaction and Feedback

```

[ ]: crosstab_comp=pd.crosstab(df['Complain'],df['Exited'],normalize='index')
crosstab_comp

```

```

[ ]: Exited          0          1
Complain
0          0.999497  0.000503
1          0.004892  0.995108

```

```
[ ]: # Ho:There is no significant effect of Complaints on Churn rates.
# Ha:There is a significant effect of Complaint on Churn rates
```

```
chi2, p, dof, expected = chi2_contingency(pd.
    ↪crosstab(df['Complain'],df['Exited']))

# Print the results of the chi-square test
print("Chi-square test statistic:", chi2)
print("P-value:", p)

# Interpret the results
if p < 0.05:
    print('Reject the Null Hypothesis')
else:
    print("Fail to Reject the Null Hypothesis")
```

Chi-square test statistic: 9907.907035880155

P-value: 0.0

Reject the Null Hypothesis

## 6 Card Usage Analysis

```
[ ]: crosstab_card=pd.crosstab(df['Card Type'],df['Exited'],normalize='index')
crosstab_card
```

```
[ ]: Exited          0          1
Card Type
DIAMOND      0.782210  0.217790
GOLD         0.807354  0.192646
PLATINUM     0.796393  0.203607
SILVER       0.798878  0.201122
```

```
[ ]: chi2, p, dof, expected = chi2_contingency(pd.crosstab(df['Card_
    ↪Type'],df['Exited']))
```

```
# Print the results of the chi-square test
print("Chi-square test statistic:", chi2)
print("P-value:", p)

# Interpret the results
if p < 0.05:
    print('Reject the Null Hypothesis')
else:
    print("Fail to Reject the Null Hypothesis")
```

Chi-square test statistic: 5.053223027060927

P-value: 0.16794112067810177

Fail to Reject the Null Hypothesis

```
[ ]: #Ho:There is no significant difference in Both the groups(Churned and not
      ↪Churned)
      # Ha:There is a significant difference in Both the groups(Churned and not
      ↪Churned)

      tstats,p_val=ttest_ind(data_0['Point Earned'],data_1['Point Earned'])
      print("tstats:",tstats)
      print("p_val:",p_val)
      if p_val<0.05:
          print('Reject Null Hypothesis')
      else:
          print('fail to reject Null Hypothesis')
```

tstats: 0.4627759848070133

p\_val: 0.6435350184288993

fail to reject Null Hypothesis

## 7 Salary Analysis

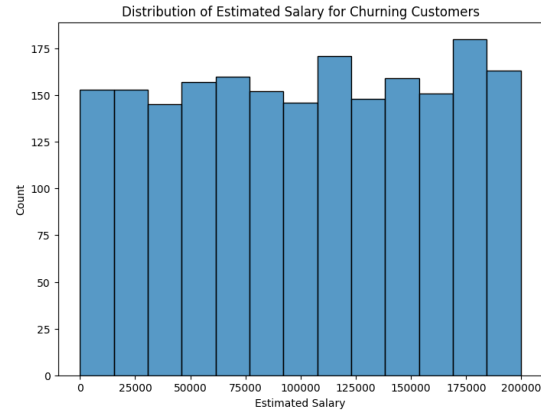
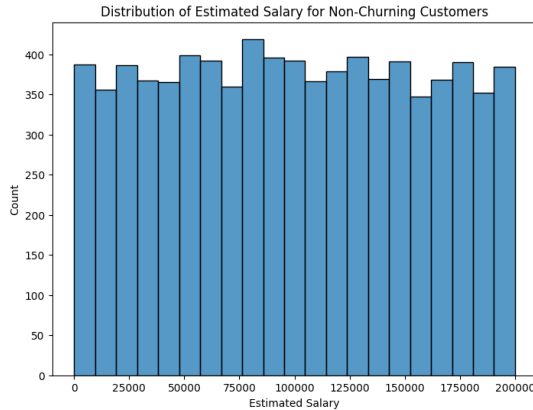
```
[ ]: df.groupby('Exited')['EstimatedSalary'].agg(['mean', 'median'])
```

```
[ ]:          mean      median
Exited
0      99726.853141   99620.355
1     101509.908783  102489.335
```

```
[ ]: plt.figure(figsize=(18, 6))
      plt.subplot(1, 2, 1)
      sns.histplot(data_0['EstimatedSalary'],label='Non-Churning')
      plt.title('Distribution of Estimated Salary for Non-Churning Customers')
      plt.xlabel('Estimated Salary')
      plt.ylabel('Count')

      plt.subplot(1, 2, 2)
      sns.histplot(data_1['EstimatedSalary'],label='Churning')
      plt.title('Distribution of Estimated Salary for Churning Customers')
      plt.xlabel('Estimated Salary')
      plt.ylabel('Count')
```

```
[ ]: Text(0, 0.5, 'Count')
```



```
[ ]: Ho:There is no significant difference in Both the groups(Churned and not_
    ↪Churned)
# Ha:Ho:There is a significant difference in Both the groups(Churned and not_
    ↪Churned)
tstats,p_val=ttest_ind(data_0['EstimatedSalary'],data_1['EstimatedSalary'])
print("tstats:",tstats)
print("p_val:",p_val)
if p_val<0.05:
    print('Reject Null Hypothesis')
else:
    print('fail to reject Null Hypothesis')
```

```
tstats: -1.2489445044833742
p_val: 0.2117146135149097
fail to reject Null Hypothesis
```

## 8 Insights:

- Customers between 40-50 age group are more likely to churn than other age group customers.
- Female customers show a slightly higher churn rate compared to male customers.
- Customers from Germany show a higher tendency to leave than France and Spain.
- Customers with fewer products (especially those with 1-2 products) are more likely to churn.
- Active members are significantly less likely to churn.
- Customers who have filed complaints are more likely to churn.
- There is no impact of having premium card types on churn rates.
- There is no impact of Points Earned on Churning rates.
- There is no impact of estimated salary on churn rates.

## 9 Recommendation

Enhance Customer Satisfaction: - Improve customer service training and responsiveness. Implement a feedback system for continuous service improvement. Offer personalized services and communication based on customer data.

Loyalty Programs: - Introduce loyalty rewards, such as points-based systems and discounts. Provide special incentives for long-tenure customers, like better interest rates and exclusive products.

Financial Advisory Services: - Offer proactive financial advice, especially to customers with lower balances. Schedule regular financial check-ins to optimize customer banking and investment strategies.

Improve Product Offering: - Expand the product portfolio to include more tailored options for different customer segments. Create bundled service packages offering multiple products at a discounted rate.

Address Customer Complaints: - Implement a robust system for efficient complaint resolution. Conduct root cause analysis of complaints to address underlying issues.

Targeted Marketing Campaigns: - Use predictive analytics to identify high-risk churn customers and target them with retention campaigns. Run customer engagement campaigns, such as newsletters, financial literacy programs, and webinars.

Monitor and Improve: - Continuously monitor key metrics like churn rate, customer satisfaction scores, and Net Promoter Score (NPS). Regularly review and iterate on retention strategies based on monitoring results.

Enhance Digital Experience: - Invest in improving the user experience of online and mobile banking platforms. Simplify the digital onboarding process and provide virtual support through chatbots and online assistance.

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